

A Technical Introduction to TensorFlow

Section 1

Losses To train and assess models, TensorFlow provides a set of loss functions (also known as cost functions). Some popular examples include mean squared error (MSE) and binary cross entropy (BCE).

Section 2

Medical GE Healthcare used TensorFlow to increase the speed and accuracy of MRIs in identifying specific body parts. Google used TensorFlow to create DermAssist, a free mobile application that allows users to take pictures of their skin and identify potential health complications. Sinovation Ventures used TensorFlow to identify and classify eye diseases from optical coherence tomography (OCT) scans.

Section 3

TensorFlow.js TensorFlow also has a library for machine learning in JavaScript. Using the provided JavaScript APIs, TensorFlow.js allows users to use either TensorFlow.js models or converted models from TensorFlow or TFLite, retrain the given models, and run on the web.

Section 4

TF.nn TensorFlow.nn is a module for executing primitive neural network operations on models. Some of these operations include variations of convolutions (1/2/3D, Atrous, depthwise), activation functions (Softmax, RELU, GELU, Sigmoid, etc.) and their variations, and other operations (max-pooling, bias-add, etc.).

Section 5

TensorFlow is a software library for machine learning and artificial intelligence. It can be used across a range of tasks, but is used mainly for training and inference of neural networks. It is one of the most popular deep learning frameworks, alongside others such as PyTorch. It is free and open-source software released under the Apache License 2.0. It was developed by the Google Brain team for Google's internal use in research and production. The initial version was released under the Apache License 2.0 in 2015. Google released an updated version, TensorFlow 2.0, in September 2019. TensorFlow can be used in a wide variety of programming languages, including Python, JavaScript, C++, and Java, facilitating its use in a range of applications in many sectors.

Section 6

Social media Twitter implemented TensorFlow to rank tweets by importance for a given user, and changed their platform to show tweets in order of this ranking. Previously,

tweets were simply shown in reverse chronological order. The photo sharing app VSCO used TensorFlow to help suggest custom filters for photos.

Section 7

TFX TensorFlow Extended (abbrev. TFX) provides numerous components to perform all the operations needed for end-to-end production. Components include loading, validating, and transforming data, tuning, training, and evaluating the machine learning model, and pushing the model itself into production.

Section 8

Eager execution TensorFlow includes an "eager execution" mode, which means that operations are evaluated immediately as opposed to being added to a computational graph which is executed later. Code executed eagerly can be examined step-by-step through a debugger, since data is augmented at each line of code rather than later in a computational graph. This execution paradigm is considered to be easier to debug because of its step by step transparency.

Section 9

Numpy Numpy is one of the most popular Python data libraries, and TensorFlow offers integration and compatibility with its data structures. Numpy NDarrays, the library's native datatype, are automatically converted to TensorFlow Tensors in TF operations; the same is also true vice versa. This allows for the two libraries to work in unison without requiring the user to write explicit data conversions. Moreover, the integration extends to memory optimization by having TF Tensors share the underlying memory representations of Numpy NDarrays whenever possible.

Section 10

TensorFlow TensorFlow serves as a core platform and library for machine learning. TensorFlow's APIs use Keras to allow users to make their own machine-learning models. In addition to building and training their model, TensorFlow can also help load the data to train the model, and deploy it using TensorFlow Serving. TensorFlow provides a stable Python Application Program Interface (API), as well as APIs without backwards compatibility guarantee for JavaScript, C++, and Java. Third-party language binding packages are also available for C#, Haskell, Julia, MATLAB, Object Pascal, R, Scala, Rust, OCaml, and Crystal. Bindings that are now archived and unsupported include Go and Swift.

Section 11

DistBelief Starting in 2011, Google Brain built DistBelief as a proprietary machine learning system based on deep learning neural networks. Its use grew rapidly across

diverse Alphabet companies in both research and commercial applications. Google assigned multiple computer scientists, including Jeff Dean, to simplify and refactor the codebase of DistBelief into a faster, more robust application-grade library, which became TensorFlow. In 2009, the team, led by Geoffrey Hinton, had implemented generalized backpropagation and other improvements, which allowed generation of neural networks with substantially higher accuracy, for instance a 25% reduction in errors in speech recognition.

Section 12

Edge TPU In July 2018, the Edge TPU was announced. Edge TPU is Google's purpose-built ASIC chip designed to run TensorFlow Lite machine learning (ML) models on small client computing devices such as smartphones known as edge computing.

Section 13

Optimizers TensorFlow offers a set of optimizers for training neural networks, including ADAM, ADAGRAD, and Stochastic Gradient Descent (SGD). When training a model, different optimizers offer different modes of parameter tuning, often affecting a model's convergence and performance.

Section 14

Education InSpace, a virtual learning platform, used TensorFlow to filter out toxic chat messages in classrooms. Liulishuo, an online English learning platform, utilized TensorFlow to create an adaptive curriculum for each student. TensorFlow was used to assess the current abilities of students and also helped decide which content to show based on those capabilities.

Section 15

In May 2016, Google announced its Tensor processing unit (TPU), an application-specific integrated circuit (ASIC, a hardware chip) built specifically for machine learning and tailored for TensorFlow. A TPU is a programmable AI accelerator designed to provide high throughput of low-precision arithmetic (e.g., 8-bit), and oriented toward using or running models rather than training them. Google announced they had been running TPUs inside their data centers for more than a year, and had found them to deliver an order of magnitude better-optimized performance per watt for machine learning. In May 2017, Google announced the second-generation, as well as the availability of the TPUs in Google Compute Engine. The second-generation TPUs deliver up to 180 teraflops of performance, and when organized into clusters of 64 TPUs, provide up to 11.5 petaflops. In May 2018, Google announced the third-generation TPUs delivering up to 420 teraflops of performance and 128 GB high bandwidth memory (HBM). Cloud TPU v3 Pods offer 100+ petaflops of performance

and 32 TB HBM. In February 2018, Google announced that they were making TPUs available in beta on the Google Cloud Platform.